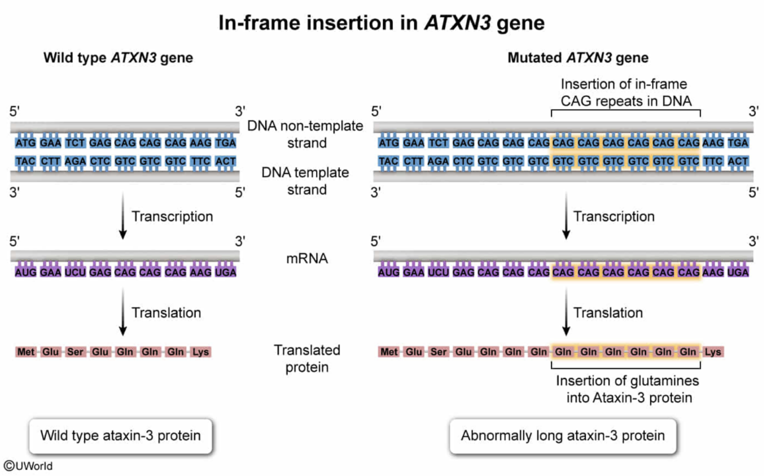
Spinocerebellar ataxia (SCA) is a disorder caused by the expansion of multiple three-nucleotide sequence repeats (CAG) within the coding region of the *ATXN3* gene, leading to a mutated version of the protein, ataxin-3. What would be the consequence of this type of mutation?

- A. It would form an abnormally long ataxin-3 protein. [65%]
- B. It would result in a shortened ataxin-3 protein. [2%]
- C. It would create an alternatively spliced version of *ATXN3* mRNA. [10%]
- D. It would shift the reading frame during translation. [21%]

Correct
65% answered correctly

Explanation:



The central dogma of biology governs the flow of information from gene to protein. Information stored in DNA is transcribed into mRNA, and the mRNA is translated into a protein by the ribosome.

During transcription, mRNA complementary to a DNA template is synthesized. Subsequently, the mRNA is processed with noncoding portions (ie, introns) removed, and coding portions (ie, **exons**) joined. A process known as alternative splicing generates multiple mRNA transcripts, and different proteins can be formed from a single gene.

The mRNA transcript is read by the ribosome in groups of three (ie, codons). The reading frame is set by the start codon, which is always AUG. During translation, the start codon is recognized by the anticodon of its corresponding tRNA. Subsequent codons are recognized by the appropriate tRNA, which delivers the correct amino acid to the growing protein.

The question states that multiple three-nucleotide repeats (CAG)

are inserted into the coding sequence of the *ATXN3* gene. Because CAG is a group of three nucleotides, this insertion maintains the **correct reading frame** and results in the repeated incorporation of the same amino acid (ie, glutamine), **forming an abnormally long ataxin-3 protein**.

(Choice B) Expansion of the CAG three-nucleotide repeat results in the inserted sequences being transcribed and translated, extending (*not* shortening) the overall length of the protein. A shortened protein would result if the inserted nucleotides contained a stop codon (ie, UAA, UAG, UGA).

(Choice C) mRNA processing (eg, splicing) happens post-transcriptionally. Insertion of CAG repeats within the DNA coding sequence should *not* affect the splicing of the mRNA because coding regions are not removed during splicing.

(Choice D) Because CAG consists of three nucleotides, the expansion of additional CAG repeats should *not* affect the overall reading frame. An insertion of this type would result in additional copies of the same amino acid being inserted into the protein without shifting the reading frame.

Educational objective:

The central dogma of biology explains how genes are expressed. DNA is transcribed into mRNA, and mRNA is translated into the final protein product by the ribosome. For mRNA to be translated into the correct protein, a ribosome must read the mRNA codons in the correct reading frame.

Time spent:0 sec
Subject:
Biology

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Molecular Biology